Akoizo Original Practice - 2027 Rotation Form 1 - Answer key

Division B | Remote Sensing | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Classify Sentinel-1 SAR as active or passive and explain the distinction.

Synthetic Sentinel-2 interpretation exercise: square pixels are 15 m on a side. A rectangular changed patch is 130 pixels east-west and 65 pixels north-south. Map north is up. A paired Sentinel-1 image was acquired through cloud. A separate Landsat image has water, vegetation and built-area classes; all dates, bands and measurements used in each question are stated.

answer: Active: the radar transmits a signal and measures its return. (2 points).

2. (2 points) Which kind of resolution describes the ground area represented by a pixel?

Synthetic Sentinel-2 interpretation exercise: square pixels are 15 m on a side. A rectangular changed patch is 130 pixels east-west and 65 pixels north-south. Map north is up. A paired Sentinel-1 image was acquired through cloud. A separate Landsat image has water, vegetation and built-area classes; all dates, bands and measurements used in each question are stated.

answer: Spatial resolution. (2 points).

3. (2 points) Does a shorter orbital period necessarily give an equally short revisit time for a specified ground point? Explain.

Synthetic Sentinel-2 interpretation exercise: square pixels are 15 m on a side. A rectangular changed patch is 130 pixels east-west and 65 pixels north-south. Map north is up. A paired Sentinel-1 image was acquired through cloud. A separate Landsat image has water, vegetation and built-area classes; all dates, bands and measurements used in each question are stated.

answer: No. Revisit depends on orbit geometry, swath and pointing as well as orbital period. (2 points).

4. (2 points) Using c=3.0e8 m/s, find the frequency of light with wavelength 6.0e-7 m.

answer: 500000000000000 Hz (2 points).

Numerical target 5e+14; absolute tolerance 1e+12. Accepted units: , Hz.

5. (2 points) Use Wien's law with b=2.90e-3 m K to find peak wavelength at 290 K.

answer: 0.00001 m (2 points).

Numerical target 1e-05; absolute tolerance 2e-08. Accepted units: , m.

6. (2 points) A radar echo takes 2.0e-5 s for its round trip. With c=3.0e8 m/s, calculate range.

answer: 3000 m (2 points).

Numerical target 3000; absolute tolerance 6. Accepted units: , m.

7. (2 points) Find the changed patch east-west ground length.

Synthetic Sentinel-2 interpretation exercise: square pixels are 15 m on a side. A rectangular changed patch is 130 pixels east-west and 65 pixels north-south. Map north is up. A paired Sentinel-1 image was acquired through cloud. A separate Landsat image has water, vegetation and built-area classes; all dates, bands and measurements used in each question are stated.

answer: 1950 m (2 points).

Numerical target 1950; absolute tolerance 3.9. Accepted units: , m.

8. (2 points) Find the changed patch north-south ground length.

Synthetic Sentinel-2 interpretation exercise: square pixels are 15 m on a side. A rectangular changed patch is 130 pixels east-west and 65 pixels north-south. Map north is up. A paired Sentinel-1 image was acquired through cloud. A separate Landsat image has water, vegetation and built-area classes; all dates, bands and measurements used in each question are stated.

answer: 975 m (2 points).

Numerical target 975; absolute tolerance 1.95. Accepted units: , m.

9. (2 points) Find its ground area.

Synthetic Sentinel-2 interpretation exercise: square pixels are 15 m on a side. A rectangular changed patch is 130 pixels east-west and 65 pixels north-south. Map north is up. A paired Sentinel-1 image was acquired through cloud. A separate Landsat image has water, vegetation and built-area classes; all dates, bands and measurements used in each question are stated.

answer: 1901250 m^2 (2 points).

Numerical target 1901250; absolute tolerance 3802.5. Accepted units: , m^2.

10. (2 points) Find its perimeter.

Synthetic Sentinel-2 interpretation exercise: square pixels are 15 m on a side. A rectangular changed patch is 130 pixels east-west and 65 pixels north-south. Map north is up. A paired Sentinel-1 image was acquired through cloud. A separate Landsat image has water, vegetation and built-area classes; all dates, bands and measurements used in each question are stated.

answer: 5850 m (2 points).

Numerical target 5850; absolute tolerance 11.7. Accepted units: , m.

11. (2 points) Cloud obscures the optical image but the SAR image shows a surface pattern. Explain why these observations differ.

Synthetic Sentinel-2 interpretation exercise: square pixels are 15 m on a side. A rectangular changed patch is 130 pixels east-west and 65 pixels north-south. Map north is up. A paired Sentinel-1 image was acquired through cloud. A separate Landsat image has water, vegetation and built-area classes; all dates, bands and measurements used in each question are stated.

answer: Visible/near-infrared optical observations can be obscured by cloud; radar can often observe through cloud. Interpret radar geometry and surface scattering before classifying the pattern. (2 points).

12. (2 points) A dark SAR patch is labeled water from brightness alone. Give one alternative explanation and a stronger check.

Synthetic Sentinel-2 interpretation exercise: square pixels are 15 m on a side. A rectangular changed patch is 130 pixels east-west and 65 pixels north-south. Map north is up. A paired Sentinel-1 image was acquired through cloud. A separate Landsat image has water, vegetation and built-area classes; all dates, bands and measurements used in each question are stated.

answer: Smooth non-water surfaces, shadow or viewing geometry can also be dark. Compare location, terrain, another date and optical/ground evidence. (2 points).

13. (2 points) A classified map labels 41 of 100 equal-area pixels as forest. Calculate forest cover percentage.

answer: 41 % (2 points).

Numerical target 41; absolute tolerance 0.082. Accepted units: , %.

14. (2 points) Forest pixels decline from 81 to 51. How many equal-area pixels of forest are lost?

answer: 30 (2 points).

Numerical target 30; absolute tolerance 0.06. Accepted units: , .

15. (2 points) Explain why comparing a wet-season image with a dry-season image can mimic land-cover change.

answer: Seasonal vegetation, water and illumination can change reflectance without permanent cover conversion; compare similar seasons and acquisition conditions. (2 points).

16. (2 points) A topographic profile rises 60 m over 1,200 m horizontal distance. Find gradient as a percentage.

answer: 5 % (2 points).

Numerical target 5; absolute tolerance 0.01. Accepted units: , %.

17. (2 points) Two contours are 20 m apart vertically. Adjacent contours are closer horizontally in area A than area B. Which area is steeper and why?

answer: A; the same elevation change occurs over a shorter horizontal distance. (2 points).

18. (2 points) A false-color display makes healthy vegetation red. Does that mean the vegetation reflects red light most strongly? Explain.

answer: No. Display colors are assigned to selected sensor bands; red display can represent near-infrared reflectance. (2 points).

19. (2 points) An ECOSTRESS thermal image shows a warmer paved area than a nearby canopy. Give an observation and a cautious interpretation.

answer: The measured surface-temperature estimate is higher over pavement; material properties, shading and evapotranspiration may contribute. Do not equate surface temperature with air temperature without evidence. (2 points).

20. (2 points) A classifier was trained using labeled examples of water, forest and pavement. Is it supervised or unsupervised? Explain.

answer: Supervised, because labeled training examples associate measurements with classes. (2 points).